# Federal Board SSC-I Examination
# Physics Model Question Paper

**National Curriculum 2022-2023 (Inclusive Scheme of Studies 2024)**

**(50% MCQs & 50% Descriptive)**

## Section - A (Marks 30)

**Time Allowed: 1.20 hours**

**Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/ overwriting is not allowed. Use black or blue ball point to fill the bubbles.** *Do not use lead pencil.*

| ROLL NUMBER | | | | | |
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| ⓪ | ⓪ | ⓪ | ⓪ | ⓪ | ⓪ |
| ① | ① | ① | ① | ① | ① |
| ② | ② | ② | ② | ② | ② |
| ③ | ③ | ③ | ③ | ③ | ③ |
| ④ | ④ | ④ | ④ | ④ | ④ |
| ⑤ | ⑤ | ⑤ | ⑤ | ⑤ | ⑤ |
| ⑥ | ⑥ | ⑥ | ⑥ | ⑥ | ⑥ |
| ⑦ | ⑦ | ⑦ | ⑦ | ⑦ | ⑦ |
| ⑧ | ⑧ | ⑧ | ⑧ | ⑧ | ⑧ |
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| Version No. | | | |
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| ⑤ | ⑤ | ⑤ | ⑤ |
| ⑥ | ⑥ | ⑥ | ⑥ |
| ⑦ | ⑦ | ⑦ | ⑦ |
| ⑧ | ⑧ | ⑧ | ⑧ |
| ⑨ | ⑨ | ⑨ | ⑨ |

**Candidate Sign. ________** **Invigilator Sign. ____________**

**Note: Fill the correct bubble for each question. Each question carries 0.5 mark.**

| S # | Question | (A) | (B) | (C) | (D) |
|---|---|---|---|---|---|
| **1.** | Which of the following is a base quantity? | Velocity | Mass | Force | Momentum |
| **2.** | The rate of change of displacement is called: | Speed | Velocity | Acceleration | Distance |
| **3.** | Which of the following forces is a non-contact force? | Friction | Tension | Magnetic force | Air resistance |
| **4.** | The tendency of an object to resist a change in its state of rest or motion is called: | Force | Weight | Inertia | Momentum |
| **5.** | Pascal's Law is applicable to: | Gases only | Solids only | Fluids | Vacuum |
| **6.** | Which of the following is a scalar quantity? | Force | Displacement | Energy | Velocity |
| **7.** | The SI unit of momentum is: | N m | kg m/s | kg m/s$^2$ | J |
| **8.** | Which statement best describes translational motion of a body? | all its parts move in one direction | rotates about a fixed axis | moves about a fixed point | Changes its shape |
| **9.** | The value of 'g' at the surface of the moon is approximately: | 9.8 m/s$^2$ | 1.6 m/s$^2$ | Infinite | 0 m/s$^2$ |
| **10.** | Two identical beakers have same amount of water. Beaker A is at 30$^o$C while beaker B is at 60$^o$C temperature. This is because, | Particles of beaker B has more internal energy | Particles of beaker A has more density | Particles of beaker A has more internal energy | Particles of beaker B has less density |

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| **11.** | A book is lifted from the floor to a shelf. It gains gravitational potential energy (P.E). Which factor is responsible for the gain in P.E of book? | Energy lost by the book as heat | Work done against gravity to raise the book | The increase in the speed of book | The increase in the weight of book |
| **12.** | Which one of the following is a renewable energy source? | Coal | Natural gas | Solar energy | Nuclear energy |
| **13.** | A group of students carries a liquid barometer while climbing a mountain. They observe that the mercury level decreases as they go higher. What is the reason for this decrease? | Lower air pressure | Stronger gravity | Lower temperature | Higher air pressure |
| **14.** | The center of gravity of a uniform rod lies at: | it's one end | its center | One-quarter from an end | outside the rod |
| **15.** | What is the purpose of using ball bearings in a machine? | they increase friction | they reduce friction | they increase mass | they reduce air resistance |
| **16.** | A file size is 2 gigabytes (GB). If expressed in megabytes (MB), what is the correct value? | 2 MB | 20 MB | 200 MB | 2000 MB |
| **17.** | Which factor determines validity of a theory? | Length of time | Experimental evidence | Popularity | Number of equations |
| **18.** | In a hydraulic press, the pressure applied on the small piston is: | decreased | transmitted equally | increased | vanished |
| **19.** | The energy stored in a compressed spring is: | Kinetic energy | Elastic potential energy | Chemical energy | Thermal energy |
| **20.** | The slope of a distance-time graph gives: | Acceleration | Force | Speed | Work |
| **21.** | A shopping cart is pushed with a force of 2 N for 5 s. What impulse is given to the cart? | 2 Ns | 5 Ns | 10 Ns | 20 Ns |
| **22.** | A factory machine has an efficiency of 30%. It means that: | 30% of the input energy is lost as heat | 30% of the input energy is converted into useful output | 70% of the input energy is converted into useful output | 40% of the input energy is converted into useful output |
| **23.** | When a pressure is applied on a surface, the force acts: | Parallel to the surface | Perpendicular to the surface | At $45^{o}$ angle to the surface | In the same direction as gravity |
| **24.** | When a bus starts suddenly, passengers fall backward due to: | Friction | Weight | Inertia | Gravitation |
| **25.** | An astronaut takes an object that weighs 200 N on Earth to an unknown planet, where it weighs 50N. If the mass of the object is | 0.4N/kg | 9.8 N/kg | 2.5 N/kg | 25 N/kg |

| | | | | | |
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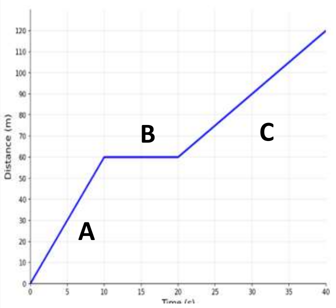
| | 20kg, what is the gravitational field strength on that planet? | | | | |
| **26.** | Like parallel forces act in: | opposite directions | same direction | at an angle of 90° to each other | at an angle of 45° to each other |
| **27.** | The state of matter found in stars is called: | Solid | Plasma | Gas | Liquid |
| **28.** | How much work is needed to lift a bag of groceries that weighs 150 N to a height of 4 m? | 300 J | 450 J | 600 J | 750 J |
| **29.** | Vector X points towards east and vector Y points towards north. In which direction does the resultant vector lie? | North -East | South-East | North | East |
| **30.** | The branch of physics which deals with study of light is: | Mechanics | Optics | Thermodynamics | Magnetism |
| **31.** | In which of the following situations are the distance traveled and the magnitude of the displacement equal? | A child moving on a swing | An object is moving along a circular path. | A car moving on a curved road | Linear motion in one direction. |
| **32.** | When an external magnetic field is removed, a paramagnetic material usually: | Remains permanently magnetized | Loses most of its induced magnetism | Becomes strongly magnetic | Repels magnetic fields strongly |
| **33.** | The distance–time graph for a cyclist moving along a straight road is shown. Which motion is displayed by the cyclist in portion A of the graph?   | At rest | Uniform speed | Non-uniform speed | Variable speed |
| **34.** | In scientific notation, 0.005 is written as: | $5 \times 10^{-3}$ | $5 \times 10^{3}$ | $5 \times 10^{-2}$ | $5 \times 10^{-4}$ |
| **35.** | A raindrop falling from a cloud reaches a constant speed before hitting the ground. At this stage, the raindrop has reached terminal velocity because: | Its acceleration becomes zero. | Its weight becomes zero. | Air resistance disappears | Gravity no longer acts on it. |
| **36.** | A spanner of length 30cm uses a force of 150N to open at tight nut. The torque is: | 10 N m | 25 N m | 35 N m | 45 N m |

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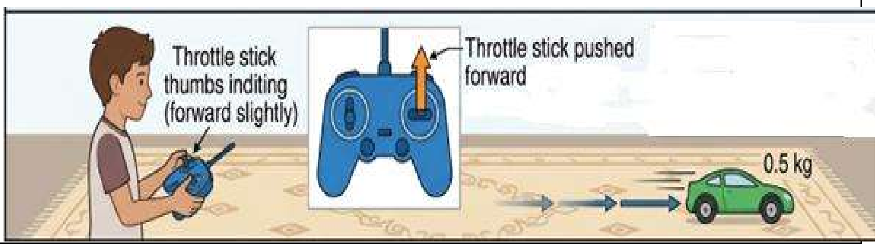
| **37.** | What is a reasonable estimate for the mass of an apple? | 2kg | 200g | 2mg | 20kg |
| **38.** | A relationship between physical quantities remains unchanged under the same conditions after many experiments. Scientists accept it as a: | Theory | Assumption | Hypothesis | Law |
| **39.** | Which of the following is a physical property that can be used to measure temperature? | mass of a liquid | volume of a liquid | mass of a substance | weight of a substance |
| **40.** | A perpetual motion machine is impossible because of: | Friction | Conservation of energy | Gravity | Air resistance |
| | **STIMULUS/SITUATION/SCENARIO BASED MCQs** | | | | |
| | **Scenario for MCQs # 41 to 44** | **Atif is playing with his remote-controlled car on a flat carpet. When he gently pushes the throttle forward, the car starts moving and gradually speeds up on straight line. Curious, he increases the force applied by the motor to observe how the motion changes.**   | | | |
| **41.** | What type of motion does the toy car exhibit? | constant velocity | constant acceleration | decreasing velocity | decreasing momentum |
| **42.** | How would the toy car's speed-time graph look like? | Straight line with negative slope | Straight line with positive slope | curved graph | Horizontal line |
| **43.** | The toy car has a mass of 0.5 kg. If it accelerates at 0.4 $m/s^2$, what is the magnitude of the force applied to it? | 0.2 N | 0.4 N | 0.8 N | 2 N |
| **44.** | He then places a small weight on the car, increasing its mass while applying the same force. What will happen to acceleration? | increases | decreases | remains same | becomes zero |
| | **Scenario for MCQs # 45 to 48** | **A worker in a scrapyard is operating a large crane fitted with an electromagnet. His job is to load and unload heavy iron scrap from trucks.** | | | |

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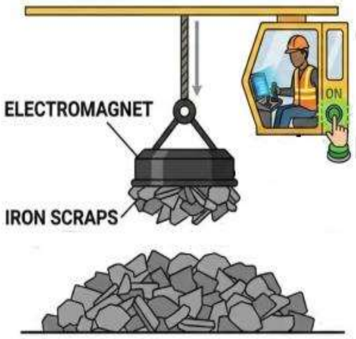
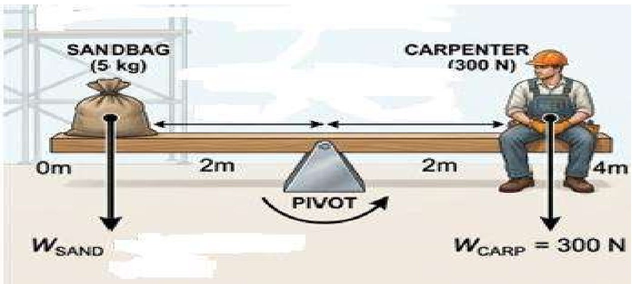
|---|---|---|---|---|---|
| | |   | | | |
| **45.** | What key idea should the worker keep in mind while operating the crane with an electromagnet? | Electro-magnet works without electricity | Magnet can be controlled (ON/OFF) | Electro-magnet only attracts plastic materials | Magnet becomes weaker as iron is attached to it |
| **46.** | Why does the crane attract only iron scraps and not plastic or wood? | Plastic and wood are lighter than iron | Plastic and wood are electrically charged | Iron is a magnetic material | Both plastic and wood block magnetic field |
| **47.** | What will happen to the crane's ability to lift and drop scrap if he replaces the electromagnet with a permanent bar magnet? | It will only attract iron pieces | It works as normal | It will attract only plastic materials | It will repel all types of materials |
| **48.** | Why is an electromagnet preferred over a permanent magnet in a scrapyard crane? | It is cheaper than a permanent magnet | It allows control over lifting and releasing scrap | It attracts all metals permanently | It does not use any electricity |
| **Scenario for MCQs # 49 to 52** | | **A carpenter is working on a construction site. He places a 5 kg sandbag on one end (left side) of a 4 m long uniform wooden beam. The beam is supported at a pivot. He then sits on the opposite side (right side) to balance the beam so that it remains horizontal.**   | | | |
| **49.** | The situation described is an example of: | Circular motion | Principle of moments | Principle of inertia | Principle of pulley |
| **50.** | Which factor does not affect the moment of a force? | Shape of the object | Magnitude of force | Distance from pivot | Direction of force |
| **51.** | If the 5 kg sandbag is now placed 1.2 m on the left side of the pivot, how far must he sit on the right side to balance the beam [$g=10ms^{-2}$]? | 0.8 m | 1.2 m | 2 m | 0.2 m |

| | | | | | |
|---|---|---|---|---|---|
| **52.** | The carpenter shifts the pivot so it is now 1 m from the sandbag (left side). What is the moment arm of carpenter? | 1 m | 2m | 3 m | 4 m |
| | **Scenario for MCQs #53 to 56** | **A space research organization is launching a new satellite into space from earth.** | | | |
| **53.** | Which expression will be used to determine the orbital velocity of satellite around the earth? | $\frac{2\pi r}{t}$ | $\frac{2\pi r}{T}$ | $\frac{2\pi T}{R}$ | $\frac{2\pi t}{r}$ |
| **54.** | The period of satellite to orbit the earth is 96.5 minutes, at a height $6.97\times 10^{6}$m from center of earth. What will be its speed in the orbit? | 7.56 km/s | 7.56 m/s | 756 km/s | 7.56 cm/s |
| **55.** | The satellite is launched through a rocket. The engine is started, hot gases are pushed downwards and the rocket lifts off the ground. Which force is directly responsible for pushing the rocket upwards? | The friction between rocket and air. | The force of engine downwards. | The force exerted by the exhaust gases pushing against the ground. | The reaction force exerted by the exhaust gases on the rocket. |
| **56.** | Which of the following explains the working of a rocket in space? | Pascal's law | Buoyant force | Newton's third law of motion | Thermal expansion of gases |
| | **Scenario for MCQs #57 to 60** | **Amir is helping his father in the garden. After digging, they need to walk across a snowy lawn to reach the tool shed. Amir is wearing his sports shoes while his father is wearing flat snowshoes.** | | | |
| **57.** | Amir steps onto the snow and his feet immediately sink deep while his father walks easily on the surface. What is the physical reason behind this? | The snowshoes increase the weight on snow. | The snowshoes decrease the weight on snow | The snowshoes increase the area of contact, decreasing pressure | The snowshoes decrease the area of contact, decreasing pressure |
| **58.** | For a constant force, what is the relation between the pressure exerted and area of contact? | direct | exponential | proportional | inverse |
| **59.** | Amir's weight is 600 N, and area of contact of his sport shoes with snow is 0.02 $m^2$. What is the pressure exerted by Amir on the snow? | 12 kPa | 30 kPa | 600 Pa | 30 Pa |
| **60.** | Amir's father has the same weight (600 N) and his snowshoes contact area is 0.20 $m^2$. How many times smaller is the pressure, he exerts as compared to Amir? | 20 times | 2 times | 5 times | 10 times |

# Federal Board SSC-I Examination
# Physics Model Question Paper

**National Curriculum 2022-2023 (Inclusive Scheme of Studies 2024)**
**(50% MCQs & 50% Descriptive)**

**Time allowed: 1 hour 40 minutes** **Total Marks: 30**

Note: Answer all parts from Section 'B' and all questions from Section 'C' on the **E-sheet**. Write your answers on the allotted/given spaces.

## SECTION – B (6x3 = 18 Marks)

**Note: Give brief answers of the following questions. All questions carry equal marks. (6x3 = 18)**

| Q# | Question | Marks | | Question | Marks |
|---|---|---|---|---|---|
| **1.** | Differentiate between base quantities and derived quantities with one example of each. | 03 | **OR** | Differentiate between mass and weight (any three points). | 03 |
| **2.** | A car starts from rest and reaches a velocity of **20m/s** in **5 seconds.** Calculate its average velocity and total distance covered by it. | 03 | **OR** | Prove that pressure of a liquid of density $\rho$ at a depth h is: $\mathbf{P = \rho g h}$. | 03 |
| **3.** | A bullet is fired from a gun. It penetrates into sand and then stops. Where does its kinetic energy is used? | 03 | **OR** | Why spring constant of steel spring is greater than rubber spring of same number of turns and same cross-sectional area? | 03 |
| **4.** | Can a body have zero velocity but non-zero acceleration? Explain with a physical example. | 1+2 | **OR** | A solenoid is having **50 cm** length and **100** turns. Calculate the magnetic field strength of solenoid if **1.2 A** current pass through it? | 03 |
| **5.** | Explain motion of rocket using law of conservation of momentum. | 03 | **OR** | How can we increase sensitivity and range of liquid-in glass thermometer? | 03 |
| **6.** | State what will happen to the rotational motion of a disc rotating freely in space, with no external moment acting on it? Provide a reason for your answer. | 1+2 | **OR** | Two identical balls are rolling on a smooth floor. Ball A has twice the momentum of ball B. Both balls are stopped by applying the same force. Which ball will take more time to come to rest? Give mathematical explanation of your reason. | 03 |

## SECTION–C (Marks 2 x 6 = 12)

| | | | | | |
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| **7.** | Suggest any three ways to magnetize and demagnetize any material. | 2+2+2 | **OR** | Describe three states of equilibrium. For each give one clear daily life example. | 2+2+2 |
| **8.** | Define acceleration. What does it tell us about the motion of an object? If the slope of a speed-time graph for a moving object is steeper, what can you explain about the motion of the object? | 2+2+2 | **OR** | State two real life situations in which energy can be converted from one form to another. Describe the mechanism by which this conversion takes place in each system. Why sometimes energy is transferred to surroundings as heat? | 2+2+2 |

# Table of Specifications (ToS) Model Paper Physics (50% MCQs Pattern) Grade IX (SSC- I)

| **Content/ Domain Areas** | **Domain A Measurement** | | **Domain B Mechanics** | | | | **Domain C Heat and Thermo-dynamics** | **Domain E Electricity And Magnetism** | **Domain G Nature of Science** | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Assessment Objectives** | **Measurements (A1-A19)** | **Kinematics (B1-B16)** | **Dynamics I (B17-B33) (B39-B42)** | **Dynamics II (B34-B38) (B43-B54) (F01-F02)** | **Pressure and Deformation in Solids (B55-B59) (B77-B86)** | **Work and Energy (B60-B76)** | **Density and Temperature (C1-C11)** | **Magnetism (E01-E15)** | **Nature of Science and Physics (G01-G08)** | **Total Marks** | **Percentage** |
| **K Knowledge** | A: 1 | A: 2 | A: 3, 4, 7, 9<br>B: 5 (f) | A: 53<br>C: 1 (s) | A: 23, 58 | A: 12, 19 | A: 27<br>B: 5 (s) | A: 46<br>C: 1 (f) | A: 30 | 25 | 27.8 % |
| **U Understanding** | A: 6, 37<br>B: 1 (f) | A: 8, 20, 29, 31<br>C: 2 (f) | A: 24, 41, 42, 44, 55, 56<br>B: 1 (s)<br>B: 6 (s) | A: 14, 15, 26, 35, 49, 50<br>B: 6 (f) | A: 5, 13, 18, 57<br>B: 3 (s) | A: 11, 22, 40<br>B: 3 (f)<br>C: 2 (s) | A: 10, 39 | A: 32, 45, 47, 48 | A: 17, 38 | 46.5 | 51.7% |
| **A Application** | A: 16, 34 | A: 33<br>B: 2 (f)<br>B: 4 (f) | A: 21, 25, 43 | A: 36, 51, 52, 54 | A: 59, 60<br>B: 2 (s) | A: 28 | --- | B: 4 (s) | --- | 18.5 | 20.5 % |
| **Total Marks** | 5.5 | 15 | 15.5 | 14.5 | 10 | 12 | 4.5 | 11.5 | 1.5 | 90 | |
| **Total Percentages** | 6.1% | 16.7% | 17.2% | 16.1 % | 11.1% | 13.3 % | 5% | 12.8 % | 1.7% | | 100 % |